Evaluation runs: chan (reliable-channel-aneris)

generated 2026-11-03 from brain-data 03b0b57cc 2026-11-03 11:21:51 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): . Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	882	conforms	0.1	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3	48	23800	conforms	0.7	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4	384	539540	conforms	7.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
5	3840	11217600	conforms	83.0	58	1 h, 250 GB	1	1	brain03, 2026-11-02
6	46080	221512004	conforms	2024.8	982	1 h, 250 GB	1	1	brain03, 2026-11-02

stateless: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	704	362	violation	0.2	2	1 h, 250 GB	363	2897	brain03, 2026-11-02
3	48	20384	15064	violation	24.4	2	1 h, 250 GB	15065	723073	brain03, 2026-11-02
4	384	543060	462840	violation	1847.5	14	1 h, 250 GB	462841	177730561	brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 5 no-seqno bag, 6 bag no-seqno, 6 bag correct, 7 correct bag, 7 bag no-seqno, 7 bag correct, 8 bag correct, 8 bag no-seqno, 9 bag no-seqno, 9 bag correct.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	8	882	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
3	48	23800	conforms	1.0	2	1 h, 250 GB	brain02, 2026-11-01
4	384	539540	conforms	8.6	6	1 h, 250 GB	brain02, 2026-11-01
5	3840	11217600	conforms	97.0	60	48 h, 250 GB	brain05, 2026-11-02
6	46080	221512004	conforms	2462.0	984	48 h, 250 GB	brain05, 2026-11-02
7	645120	4226823920	conforms	53948.0	19216	48 h, 250 GB	brain04, 2026-11-02

stateful: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	8	704	362	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-01
3	48	20384	15064	violation	1.1	2	1 h, 250 GB	brain02, 2026-11-01
4	384	543060	462840	violation	9.0	8	1 h, 250 GB	brain02, 2026-11-01
5	3840	14061696	12864192	violation	129.7	139	48 h, 250 GB	brain05, 2026-11-02
6	46080	362542466	344704646	violation	4067.0	2208	48 h, 250 GB	brain05, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 5 correct bag, 6 no-seqno bag, 7 correct bag, 7 bag no-seqno, 8 bag correct, 8 bag no-seqno, 9 bag no-seqno, 9 bag correct.

left out by the table rules (ok): 5 no-seqno bag, 6 correct bag.

left out by the table rules (timeout): 6 no-seqno bag, 7 correct bag.

skipped: a smaller size had reached its limit: 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	23800	conforms	1.2	4	10 min, 16 GB	0.1	1.1	0.0	in-memory	brain05, 2026-11-02
4	384	539540	conforms	8.8	6	10 min, 16 GB	0.1	8.7	0.0	in-memory	brain05, 2026-11-02
5	3840	11217600	conforms	102.1	54	1 h, 250 GB	0.4	101.7	0.0	merge	brain05, 2026-11-03
6	46080	221512004	conforms	3465.8	978	1 h, 250 GB	1.0	3464.6	0.1	merge	brain05, 2026-11-03

offline: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	20384	87	violation	1.0	4	10 min, 16 GB	0.1	0.9	0.0	in-memory	brain05, 2026-11-02
4	384	543060	1092	violation	10.9	22	10 min, 16 GB	0.1	10.8	0.0	in-memory	brain05, 2026-11-02
5	3840	14061696	14413	violation	122.9	74	1 h, 250 GB	0.4	122.1	0.4	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 6 no-seqno bag, 7 bag no-seqno, 7 correct bag, 7 bag correct.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 correct bag, 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8		conforms	0.1	4	10 min, 250 GB	0.1	0.1		in-memory	brain01, 2026-11-03
3	48		conforms	0.1	4	10 min, 250 GB	0.1	0.1		in-memory	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8		6	violation	0.1	4	10 min, 250 GB	0.1	0.1		in-memory	brain01, 2026-11-03
3	48		87	violation	0.1	4	10 min, 250 GB	0.1	0.1		in-memory	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno.tfspec-jmcimpl, 5 bag correct.tfspec-jmcimpl, 6 bag correct.tfspec-jmcimpl, 6 bag no-seqno.tfspec-jmcimpl, 7 bag no-seqno.tfspec-jmcimpl, 7 bag correct.tfspec-jmcimpl.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		882	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
3		23800	conforms	0.1	4	10 min, 250 GB		1.1	0.1	in-memory	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		704	6	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
3		20384	87	violation	0.1	4	10 min, 250 GB		0.9	0.1	in-memory	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag correct.jmcspec-tfimpl, 5 bag no-seqno.jmcspec-tfimpl, 6 bag correct.jmcspec-tfimpl, 6 bag no-seqno.jmcspec-tfimpl, 7 bag correct.jmcspec-tfimpl, 7 bag no-seqno.jmcspec-tfimpl.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	882	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	48	23800	conforms	1.3	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	704	362	violation	0.1	4	1 h, 250 GB	11	brain01, 2026-11-02
3	48	20384	15064	violation	1.2	4	1 h, 250 GB	1	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag no-seqno, 7 bag correct.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		882	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
3		23800	conforms	0.1	4	10 min, 250 GB		1.1	0.1	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		704	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
3		20384	conforms	0.1	4	10 min, 250 GB		0.9	0.1	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		704	6	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
3		20384	87	violation	0.1	4	10 min, 250 GB		0.9	0.1	in-memory	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno.jimpl-tfimpl, 5 bag correct.jimpl-tfimpl, 5 bag no-seqno.vs-correct.jimpl-tfimpl, 6 bag correct.jimpl-tfimpl, 6 bag no-seqno.vs-correct.jimpl-tfimpl, 6 bag no-seqno.jimpl-tfimpl.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882		conforms	0.2	4	10 min, 250 GB	0.1		0.2	in-memory	brain01, 2026-11-03
3	23800		conforms	0.1	4	10 min, 250 GB	1.1		0.1	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
3	20384		conforms	0.2	4	10 min, 250 GB	0.9		0.2	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882		6	violation	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
3	23800		87	violation	0.2	4	10 min, 250 GB	1.1		0.2	in-memory	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 bag correct.tfimpl-jimpl, 5 bag no-seqno.vs-correct.tfimpl-jimpl, 5 bag no-seqno.tfimpl-jimpl, 6 bag no-seqno.tfimpl-jimpl, 6 bag correct.tfimpl-jimpl, 6 bag no-seqno.vs-correct.tfimpl-jimpl.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		882	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
3		23800	conforms	0.3	4	10 min, 250 GB		1.1	0.3	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		704	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
3		20384	conforms	0.1	4	10 min, 250 GB		0.9	0.1	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2		704	6	violation	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
3		20384	87	violation	0.1	4	10 min, 250 GB		0.9	0.1	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno.jimpl-tfimpl, 5 bag correct.jimpl-tfimpl, 5 bag no-seqno.vs-correct.jimpl-tfimpl, 6 bag correct.jimpl-tfimpl, 6 bag no-seqno.vs-correct.jimpl-tfimpl, 6 bag no-seqno.jimpl-tfimpl.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
3	23800		conforms	0.1	4	10 min, 250 GB	1.1		0.1	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
3	20384		conforms	0.0	4	10 min, 250 GB	0.9		0.0	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882		6	violation	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
3	23800		87	violation	0.2	4	10 min, 250 GB	1.1		0.2	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 bag correct.tfimpl-jimpl, 5 bag no-seqno.vs-correct.tfimpl-jimpl, 5 bag no-seqno.tfimpl-jimpl, 6 bag no-seqno.tfimpl-jimpl, 6 bag correct.tfimpl-jimpl, 6 bag no-seqno.vs-correct.tfimpl-jimpl.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	9.8	280	1 h, 250 GB		brain01, 2026-11-03
3	23800	23800	conforms	63.3	5055	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	9.1	236	1 h, 250 GB		brain01, 2026-11-03
3	20384	20384	conforms	56.8	4930	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	9.9	236	1 h, 250 GB	95	brain01, 2026-11-03
3	23800	20384	15064	violation	56.5	4658	1 h, 250 GB	142	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno.vs-correct.w16, 5 bag no-seqno.w16, 5 bag correct.w16, 6 bag no-seqno.w16, 6 bag no-seqno.vs-correct.w16, 6 bag correct.w16.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	23800	23800	conforms	1.4	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	20384	20384	conforms	1.4	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	0.1	4	1 h, 250 GB	10	brain01, 2026-11-02
3	23800	20384	15064	violation	1.1	4	1 h, 250 GB	7	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag correct, 6 bag no-seqno.